

THE LATTICE OF TORSION PAIRS AND THE ZIEGLER SPECTRUM

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Plan:

§1 The lattice of torsion pairs

§2 The Ziegler spectrum

§3 Costing theory

§4 Mutation

§5 Semistable torsion pairs

§1 THE LATTICE OF TORSION PAIRS

A fin. dimensional algebra over a field k

$\text{Mod } A$ right A -modules

$\text{mod } A$ fin. generated A -modules

$$\mathcal{D}(A) := \mathcal{D}(\text{Mod } A)$$

$\mathcal{D}(A)^c$ compact objects

DEF. Two subcategories t, f of $\text{mod } A$ form a **torsion pair** in $\text{mod } A$ if

- $\text{Hom}_A(t, f) = 0$
- $\forall M \in \text{mod } A \exists$ short exact sequence

$$0 \rightarrow T \longrightarrow M \longrightarrow F \longrightarrow 0$$

$$\quad \quad \quad \cap \quad \quad \quad \cap$$

$$\quad \quad \quad t \quad \quad \quad f$$

Set $(u, v) \leq (t, f)$ if $u \leq t$

$\leadsto$ a complete lattice $(\text{tors } A, \leq)$

EXAMPLES

(1) $C \in \text{Mod } A$

$$t_C := \{ M \in \text{mod } A \mid \text{Hom}_A(M, C) = 0 \}$$

$F(C)$ smallest torsionfree class containing C

$\Rightarrow$

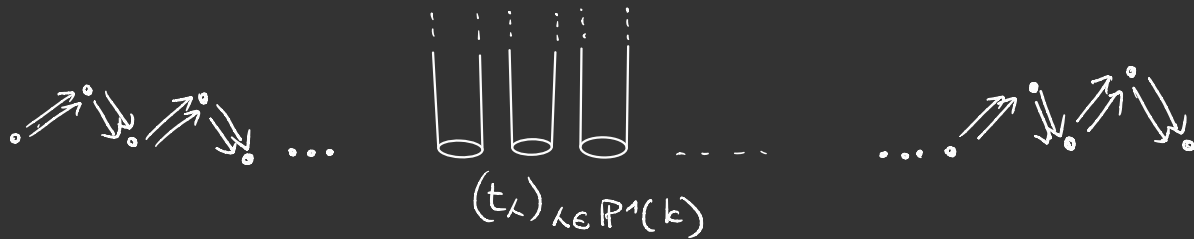
$(t_C, F(C))$ torsion pair *cogenerated* by C

Dually

$(T(C), f_C)$ torsion pair *generated* by C .

$$(2) A = k (\bullet \Longrightarrow \bullet)$$

ind A



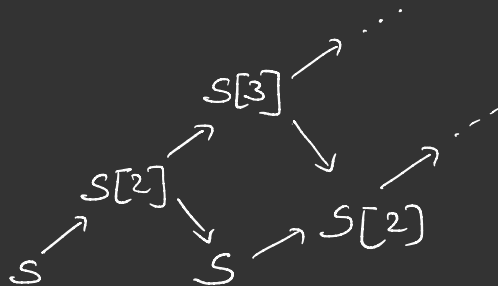
• tubes $(t_k)_{k \in P^1(k)}$

$$S : k \xrightarrow[\lambda]{1} k, \lambda \in k$$

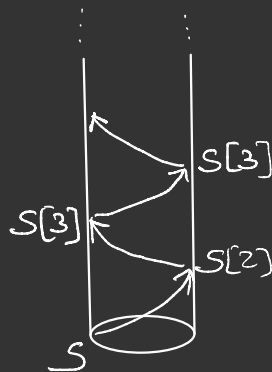
or

$$S : k \xrightarrow[1]{0} k$$

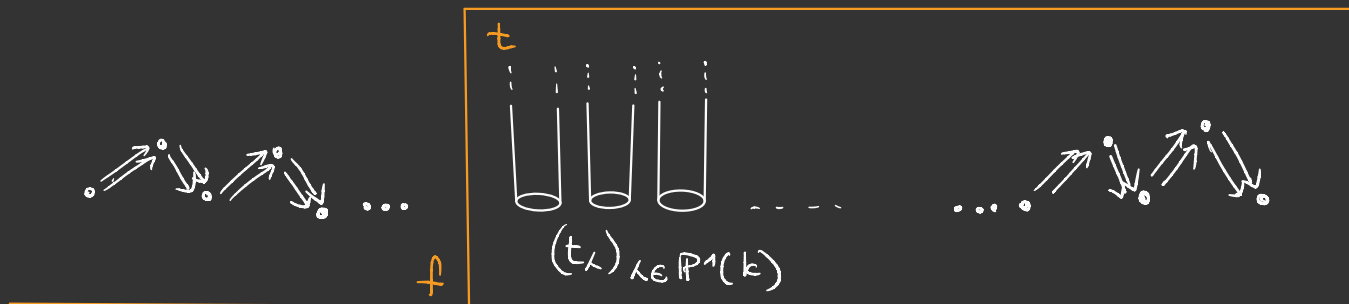
$\rightsquigarrow$



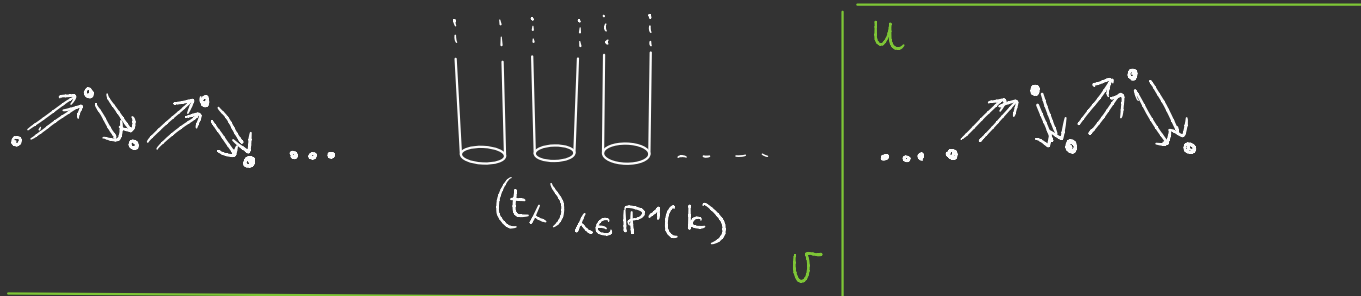
$\rightsquigarrow t_k$



(t, f)

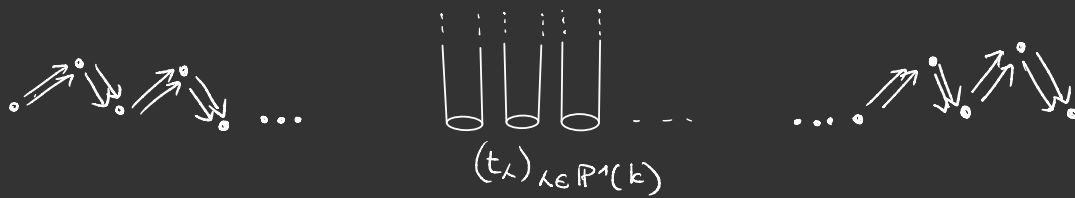


(u, v)

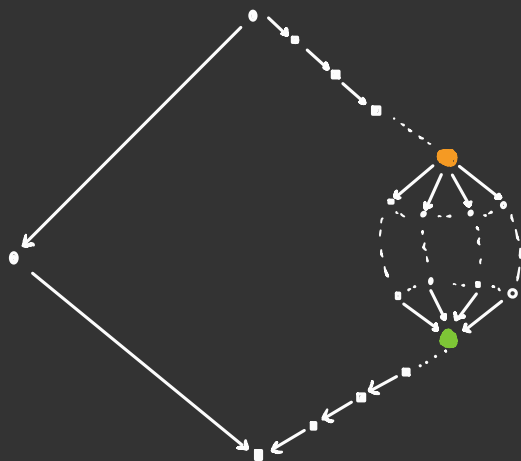


$$(u, v) \leq (t, f)$$

ind A



tors A



§2 THE ZIEGLER SPECTRUM

- DEF. • A pure triangle is a triangle $X \xrightarrow{f} Y \xrightarrow{g} Z \longrightarrow$ in $\mathcal{D}(A)$ such that every $F \in \mathcal{D}(A)^c$ yields an exact sequence
- $$0 \rightarrow \operatorname{Hom}(F, X) \rightarrow \operatorname{Hom}(F, Y) \rightarrow \operatorname{Hom}(F, Z) \rightarrow 0 \quad \text{in } \mathbf{Ab}.$$

Then f is a pure monomorphism, g a pure epimorphism.

- $X \in \mathcal{D}(A)$ is pure-injective if every pure mono $X \rightarrow Y$ splits

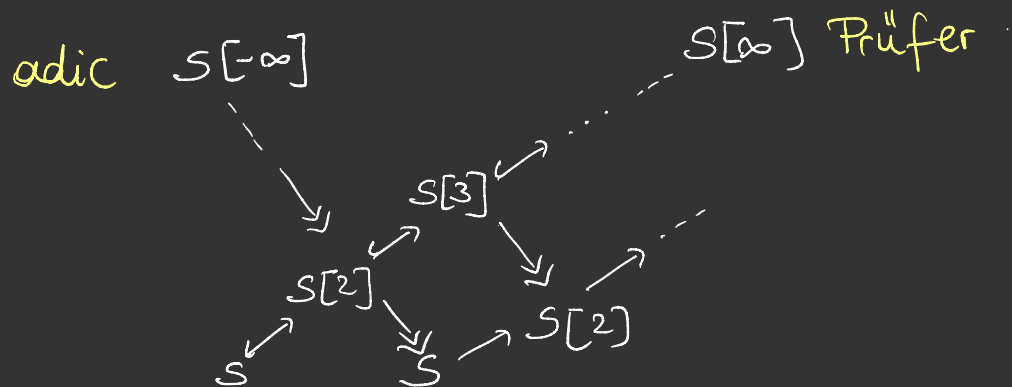
- $\mathcal{C} \subset \mathcal{D}(A)$ is **definable** if it is closed under Π , pure subobjects and pure-epimorphic images

- The **Ziegler spectrum** $\text{Zg } \mathcal{D}(A)$:
topological space

- points : indec. pure-inj. / $\cong$

- closed sets : $\mathcal{C} \cap \text{Zg } \mathcal{D}(A)$, $\mathcal{C}$ definable

EXAMPLE $A = k(\cdot \implies \cdot)$



$$G: k(x) \xrightleftharpoons[x]{1} k(x) \quad \text{generic}$$

$\{G\}, \{G, S[\infty]\}, \{G, S[-\infty]\}$ closed sets

§3 COSILTING THEORY

$$K^{[0,1]}(\text{Inj } A) := \left\{ G \in D(A) \mid G \cong \dots 0 \rightarrow I_0 \xrightarrow{\quad} I_1 \rightarrow 0 \rightarrow \dots \right\}$$

$\underbrace{\quad}_{\text{injective}}$

DEF. $G \in K^{[0,1]}(\text{Inj } A)$ is a (2-term) cosilting cpx. if

- $\text{Hom}_{D(A)}(G^I, G[1]) = 0 \quad \forall \text{ sets } I$
- $K^b(\text{Inj } A) = \text{thick} \langle \text{Prod } G \rangle$

Two cosilting cpxs G, G' are equivalent if $\text{Prod } G = \text{Prod } G'$

$$2\text{-Cosilt}(A) := 2\text{-term cosilting cpxs} / \cong$$

Facts:

(1) $\mathcal{G}$ is pure-injective [Bazzoni, Marks-Vitória]

(2) $\mathcal{N} := \text{Prod } \mathcal{G} \cap \text{Zg } \mathcal{D}(A)$ closed set in $\text{Zg } \mathcal{D}(A)$

(3) $\mathcal{N}$ rigid set, i.e.

$$\text{Hom}(\nu, \mu[1]) = 0 \quad \forall \nu, \mu \in \mathcal{N}$$

maximal among all rigid sets in

$$\text{Zg}^{[0,1]} \mathcal{D}(A) := K^{[0,1]}(\text{Inj } A) \cap \text{Zg } \mathcal{D}(A)$$

$$\text{Max Rigid}(A) := \text{max. rigid sets in } \text{Zg}^{[0,1]} \mathcal{D}(A)$$

THM 1 [A-Laking-Štornicek-Vitória, ALS]

$\exists$ bijections

$$\text{tors } A \xleftrightarrow{1-1} 2\text{-Cosilt } A \xleftrightarrow{1-1} \text{Max Rigid } A$$

$$(t_c, \mathbb{F}(C)) \longleftrightarrow G \longmapsto \mathcal{N}$$

$$C := H^0(G)$$

§4 MUTATION

DEF. $\mathcal{L}, \mathcal{R} \in \text{Max Rigid}(A)$

$\mathcal{R}$ is a right mutation of $\mathcal{L}$ ($\mathcal{L}$ left mutation of $\mathcal{R}$)

if there is a bijection $\mathcal{L} \xrightarrow{m} \mathcal{R}$ st.

- $m|_{\mathcal{L} \cap \mathcal{R}} = \text{id}$

- $\lambda \in \mathcal{L} \setminus \mathcal{R}$ is taken by m to $\rho := m(\lambda)$

given by an exchange triangle

$$\rho \xrightarrow[\text{Prod}(\mathcal{L} \cap \mathcal{R})\text{-appr.}]{\text{min. left}} \varepsilon \xrightarrow[\text{Prod}(\mathcal{L} \cap \mathcal{R})\text{-approx.}]{\text{min. right}} \lambda \longrightarrow \rho \quad [1] \text{ in } \mathcal{D}(A)$$

If $|\mathcal{L} \setminus \mathcal{R}| = 1$, irreducible mutation

DEF. An interval $[u, t]$ in $\text{tors } A$ formed by

torsion pairs $(u, v) \leq (t, f)$ is wide

if $W := t \cap v$ is wide in $\text{mod } A$ (ie. closed

under kernels, cokernels and extensions).

FACTS:

(1) [Ringel] $\exists$ set $\mathcal{B} \subseteq \text{mod } A$ of Hom-orthogonal bricks (ie. $\text{End } B$ skew-field $\forall B \in \mathcal{B}$)

sh. $\mathcal{W} = \{ \text{modules with finite } \mathcal{B}\text{-filtration} \}$

(2) [DIRRT, BC²]

$$\begin{pmatrix} t & f \\ u & v \end{pmatrix} \quad \mathcal{W} = \text{filt}(\mathcal{B})$$

$$t = u \vee T(\mathcal{B})$$

(3) [ALŠV]

$[u, t]$ wide $\Leftrightarrow \mathcal{N}_t$ is mutation of $\mathcal{N}_u$

$\exists$ arrow $\begin{array}{c} u \\ \downarrow \\ t \end{array} \Leftrightarrow \mathcal{N}_t$ is an irred. mutation of $\mathcal{N}_u$

THM 2 [ALS] There is a bijection

wide intervals
in $\text{tors} A$



Closed rigid sets
in $\text{Zg}^{[0,1]} \mathcal{D}(A)$

$[u, t]$



$\mathcal{N}_u \cap \mathcal{N}_t$

$[u_M, t_M]$

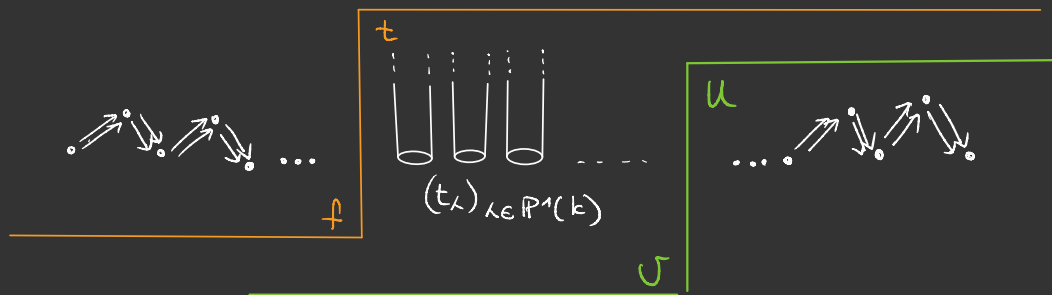


$\mathcal{M}$

"

$\{t \in \text{tors} A \mid \mathcal{M} \subseteq \mathcal{N}_t\}$

EXAMPLE $A = k(\cdot \implies \cdot)$



$[u, t]$ wide interval

$$\mathcal{N}_u = \{G\} \cup \{\text{Prüfer}\}$$

$$\mathcal{N}_t = \{G\} \cup \{\text{adics}\}$$

↕ mutation

$$\rightsquigarrow \mathcal{N}_u \cap \mathcal{N}_t = \{G\}$$

closed rigid set

§5 SEMISTABLE TORSION PAIRS

$\Theta \in K_0(\text{proj } A)$ a g -vector

$\Rightarrow$

- $\Theta = [P_0] - [P_1]$, P_0, P_1 no common summands, unique up to iso
- $\text{Hom}(\Theta) := \text{Hom}_A(P_1, P_0)$
- We regard Θ as a linear form
 $\langle \Theta, - \rangle : K_0(\text{mod } A) \otimes \mathbb{R} \longrightarrow \mathbb{R}$

- semistable torsion pairs

$$t_\theta := \{X \in \text{mod } A \mid \theta(X') > 0 \quad \forall X \twoheadrightarrow X' \neq 0\}$$

$$\overline{t}_\theta := \{X \in \text{mod } A \mid \theta(X') \geq 0 \quad \forall X \twoheadrightarrow X' \neq 0\}$$

$$f_\theta := \{X \in \text{mod } A \mid \theta(X') < 0 \quad \forall 0 \neq X' \hookrightarrow X\}$$

$$\overline{f}_\theta := \{X \in \text{mod } A \mid \theta(X') \leq 0 \quad \forall 0 \neq X' \hookrightarrow X\}$$

$$\mathcal{W}_\theta := \overline{t}_\theta \cap \overline{f}_\theta = \{\theta\text{-semistable modules}\}$$

$\rightsquigarrow$ wide interval $[t_\theta, \overline{t}_\theta]$ in $\text{tors } A$

Assume Θ is generically indecomposable (all elements on a dense open subset of $\text{Hom}(\Theta)$ are indecomposable)

[w.l.o.g. by Derksen-Fei 2015]

and A is tame.

Then

(i) Θ is rigid, or

(ii) there is a dense open subset $U \subseteq \text{Hom}(\Theta)$ st.

$$\{\text{Colker } \Theta \mid \Theta \in U\}$$

is a one-parameter family of bricks

[Planavoula-Yurkusa 2023, Geiss-Labardini-Tragoso-Schröer 2022]

EXAMPLE $A = k(\dot{} \implies \dot{})$

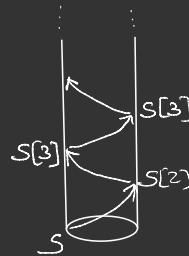
$$\theta = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

• wide interval $[t_\theta, \overline{t}_\theta] = [u, t]$

• $\text{Hom}(\theta) = \text{Hom}(P_1, P_2)$

$$0 \rightarrow P_1 \xrightarrow{G} P_2 \rightarrow S \rightarrow 0$$

where S at mouth of tube



$\rightsquigarrow (S_k)_{k \in \mathbb{P}^1(k)}$ one-parameter family of bricks

THM 3 [A-Laking-Pfeifer]

The closed rigid set $\mathcal{M}$ associated to $[t_\theta, \overline{t_\theta}]$:

in case (i)

$$\mathcal{M} = \{\mu\} \quad , \quad \mu \in k^{[0,1]}(\text{inj} A) \text{ consisting}$$

[Asai - Yama 2021]

in case ($\bar{u}$)

- there is a family of tubes and a generic module G st.

$$\mathcal{M} = \{\mu_G\}$$

- $\text{tors} A$

